Big Data Analytics (TP03) Report

Nova Super-App Dashboard Analysis

Transaction Analytics & Optimization Team

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1 Introduction & Overview

This report documents the implementation and findings of an interactive Power BI dashboard designed to process and analyze massive transaction logs from the Nova Super-App ecosystem (`nova_interactions_massive.csv`). The core objective is to evaluate overall financial performance, monitor platform transaction throughput, track system success rates, and identify regional usage trends across various core services.

2 Data Preparation & Optimization (Power Query & DAX)

To handle the scale of the dataset efficiently within system memory constraints, specific preprocessing steps were executed:

- **Schema Optimization:** Non-essential, high-cardinality fields such as `Platform_User_Agent` were dropped during ingestion to optimize memory usage and speed up cross-filtering queries.
- **Type Formatting & Null Handling:** Explicit data types were assigned to monetary fields (`Amount_USD`), timestamps (`Timestamp`), and categorical values (`Status`, `Category`).
- **Geospatial Memory Handling:** To resolve map memory rendering limitations, a targeted DAX table (`Top_50_GPS`) was generated using `TOPN` sampling to retain rendering stability while providing location context.

3 Core DAX Measures

- **Total Revenue:** Sum of overall transaction amounts formatted strictly as currency ($).
- **Total Transactions:** Count of all individual interaction records logged.
- **Success Rate & Successful Transactions:** Filtered counts isolating completed interactions to monitor operational uptime.

4 Dashboard Architecture

The Power BI visual canvas is organized into the following key components:

1. **KPI Cards:** Top-level metrics displaying *Total Revenue* and *Total Transactions*.

2. **Bar Chart:** Breakdown of *Total Revenue by Category*.
3. **Column Chart:** Distribution of *Total Transactions by Status*.
4. **Line Chart:** Temporal trends showing *Total Transactions by Month*.
5. **Slicers:** Interactive filtering controls for *Status* and *Category*.

5 Conclusion & Executive Summary

The Nova Super-App interactive Power BI dashboard provides a centralized view of multi-million transaction logs to evaluate both financial and operational health. The KPI cards establish baseline platform metrics by highlighting overall transaction volume and cumulative revenue performance. The revenue bar chart reveals clear differences in financial contribution across distinct service categories, helping isolate high-value user behavior. Operational reliability remains strong, as evidenced by the status distribution visual showing high completion rates alongside minimal system errors. Furthermore, temporal monthly trend analysis confirms sustained user engagement over time, while integrated status and category slicers allow stakeholders to conduct quick cross-sectional root cause analyses.